

Fig. 1: Circuit for digitally adjustable dancing lights

at the output of IC 4066. The selected resistor changes the time period of the 555 timer circuit, whose 'on' and 'off' timings are given by:

$$t_{on} = 0.693(R9 + R_x)C3$$

$$t_{off} = 0.693R_xC3$$

In the present design, $R9 = 1$ kilo-ohm, $C3 = 47 \mu F$ and R_x varies according to the output of IC1 (CD4017), which changes by pressing the switch S2 one by one.

When

$R_x = R5 = 5.6$ kilo-ohms, $t_{on} = 0.214969$ second, $t_{off} = 0.182398$ second

$R_x = R6 = 10$ kilo-ohms, $t_{on} = 0.358281$ second, $t_{off} = 0.325710$ second

$R_x = R7 = 22$ kilo-ohms, $t_{on} = 0.749133$

second, $t_{off} = 0.716562$ second

$R_x = R8 = 33$ kilo-ohms, $t_{on} = 1.107414$

second, $t_{off} = 1.074843$ second

Whenever you momentarily press

switch S2, the output of the decade counter advances to select a higher value of resistor R_x in IC 4066, which changes the switching time of the astable multivibrator. LED1 and LED2 indicate switching 'off' and 'on', respectively, of the multivibrator.

Assemble the circuit on a general-purpose PCB and enclose in a suitable case. Fix LEDs and switches on the front panel. Connect the optional circuit shown in Fig. 2 if higher intensity of light is needed. ●

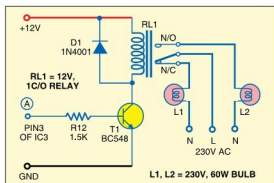


Fig. 2: Optional circuit

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